JAGSPIRE AI PRODUCT DEVELOPMENT TEAM – SPRINT 1

TEAM RESPONSIBILITIES

Intern 1 – Backend Engineer

Primary Ownership: FastAPI + Backend

Tasks:

- Set up FastAPI project architecture
- Authentication APIs
- Case Management APIs
- Evidence APIs
- File Upload APIs
- Analysis Status APIs
- Error Handling
- API Documentation

Expected Output: Working backend APIs for the product.

Intern 2 – Frontend Developer

Primary Ownership: React Dashboard

Tasks:

- React + TypeScript setup
- Login page
- Dashboard
- Case Management UI
- Case Details page
- Evidence Upload UI
- Evidence Explorer
- API Integration
- Loading & Error States

Expected Output: Functional investigator dashboard connected with backend.

Intern 3 – AI/ML Engineer

Primary Ownership: AI Investigation Engine

Tasks:

- Define AI input/output structure
- Evidence summarization
- Entity extraction
- Suspicious activity classification
- AI findings generation
- Investigation Q&A architecture
- Prompt management
- AI output validation

Expected Output: AI Analysis Engine V1.

⚠️ Do not build only a chatbot. AI must work with actual investigation/evidence data.

Intern 4 – Digital Forensics Engineer

Primary Ownership: Evidence Processing

Tasks:

- Evidence processing service
- SHA-256 hashing
- File type identification
- Metadata extraction
- EXIF extraction for supported images
- Document metadata extraction
- Basic log parsing
- Timestamp extraction
- Timeline event generation

Expected Output: Functional Forensic Evidence Processing Engine.

Intern 5 – Cyber Security Engineer

Primary Ownership: IOC & Threat Analysis

Tasks:

- IP extraction
- Domain extraction
- URL extraction
- Email extraction

- Hash extraction
- IOC normalization
- Basic threat/risk scoring
- Suspicious indicator rules
- MITRE ATT&CK mapping where applicable

Expected Output: IOC & Threat Analysis Module.

Intern 6 – Database/Data Engineer

Primary Ownership: Investigation Data Layer

Initial database entities:

- Users
- Cases
- Evidence
- Evidence Hashes
- Events
- Entities
- IOCs
- Relationships
- Timeline Events
- AI Findings
- Reports
- Audit Logs

Tasks:

- PostgreSQL schema
- SQLAlchemy models
- Relationships
- Migrations
- Indexing
- Search
- Data validation

Expected Output: Stable and integration-ready database layer.

Intern 7 – QA & DevOps Engineer

Primary Ownership: Testing + Deployment

Tasks:

- Docker setup
- Development environment
- GitHub Actions / CI
- API testing
- Integration testing
- UI testing
- Bug tracking
- Application logging
- Staging deployment

Expected Output: Tested and deployable product environment.

SPRINT 1

7 September – 13 September

Sprint Objective

By the end of this sprint, the following complete workflow should work:

Login → Create Case → Upload Evidence → Calculate Hash → Extract Metadata → Store in Database → Display Evidence in Dashboard

This is the **minimum expected outcome** of Sprint 1.

INDIVIDUAL DEADLINES

By 9 September

Intern 6

- Initial PostgreSQL schema
- SQLAlchemy models
- Migration setup

By 11 September

Intern 1

- Authentication + Case + Evidence APIs

Intern 2

- Login + Dashboard + Case + Evidence UI

Intern 4

- Hashing + File Identification + Metadata Extraction

By 12 September

Intern 3

- AI service structure + initial evidence analysis prototype

Intern 5

- IOC extraction + normalization + basic risk scoring

Intern 7

- Docker + Testing framework + CI/CD setup

13 September

Integration + Testing + Sprint Demo

All modules must be integrated and demonstrated together.

GITHUB RULES

From this phase onward:

No direct push to main

Workflow: **Issue → Feature Branch → Development → Testing → Pull Request → Review → Merge**

Use meaningful commits such as:

feat: add evidence hashing

fix: handle invalid file upload

test: add evidence API tests

docs: update API documentation

DAILY UPDATE

Every working day, post:

Name:
Yesterday:
Today:
Blocker:
GitHub/PR:

Example:

Name: XYZ
Yesterday: Completed evidence hashing
Today: Integrating metadata extraction
Blocker: EXIF extraction issue with one file format
GitHub/PR: #17

IMPORTANT

This phase is different from the previous task-based phase.

Your responsibility is not simply to **complete the assignment.**

Your responsibility is to **build, test, integrate and deliver a working part of the product.**

A module that works independently but does not integrate with the product will **not be considered complete**.

AI tools may be used for learning and development assistance, but you must understand, test and be able to explain the code you submit.

FINAL SPRINT GOAL

By **13 September**, we should be able to demonstrate:

Login → Create Investigation → Upload Evidence → Hash → Metadata → Database → Dashboard

Let's move from **learning individual concepts to building a real product as one engineering team.**